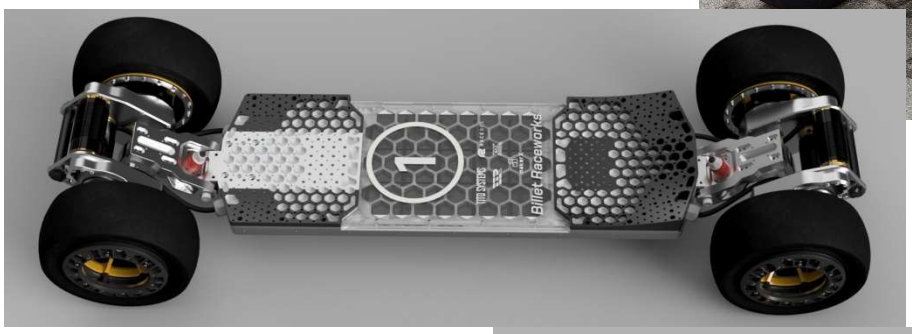
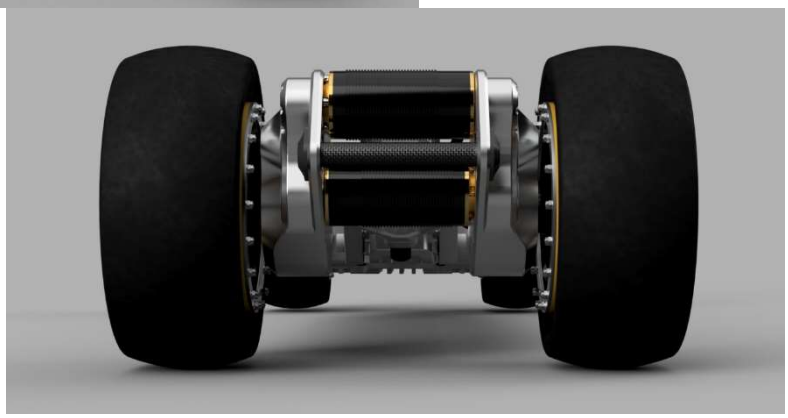
40kW racing e-skateboard project

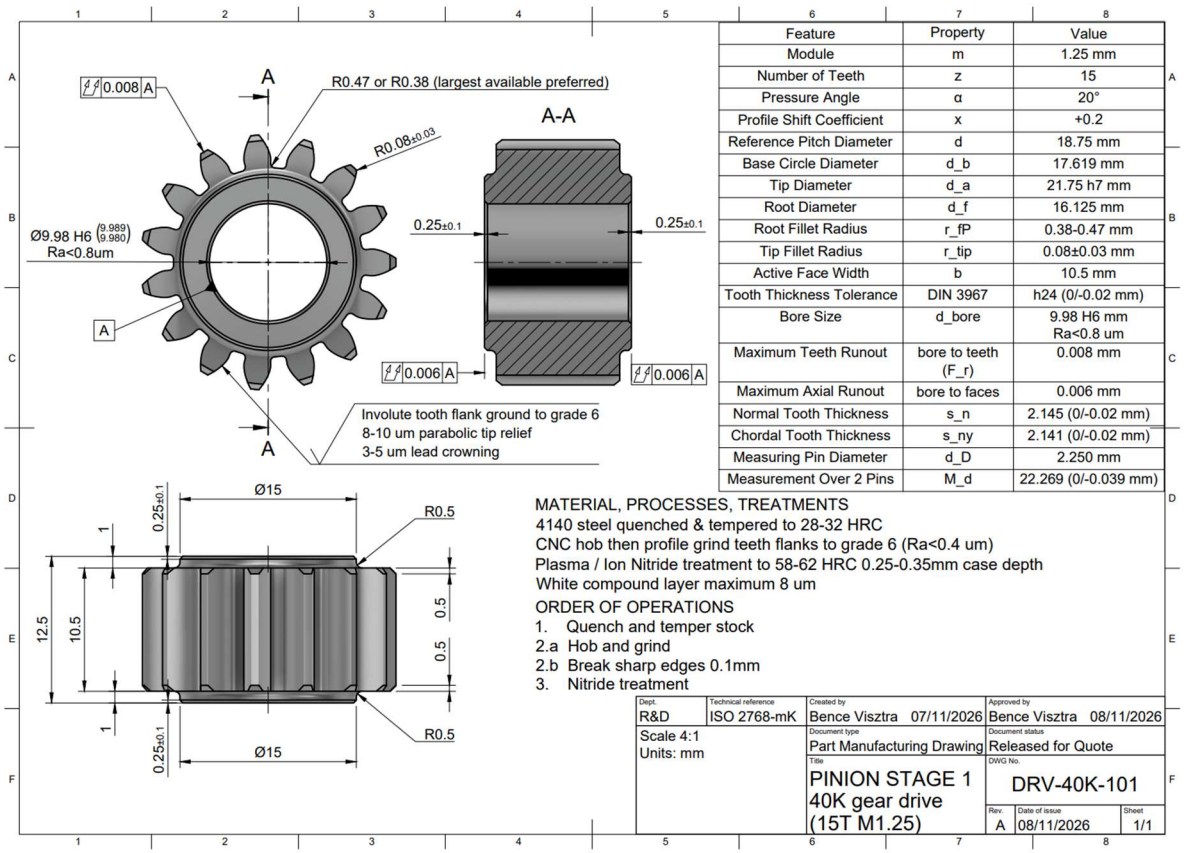
CNC milled aluminium chassis produced in-house by myself – it is production ready, including CAD and CAM for milled parts, drawings for sheet metal parts.





Custom designed double stage gear drive in the making for inrunner BLDC motors, ~40k RPM ~7 Nm ~17kW mechanical power each. Precision gearset drawings are on schedule to be sent out to my Chinese supplier by end of month. Housing will be milled in-house.





Feature	Property	Value
Module	m	1.25 mm
Number of Teeth	z	15
Pressure Angle	α	20°
Profile Shift Coefficient	x	+0.2
Reference Pitch Diameter	d	18.75 mm
Base Circle Diameter	d_b	17.619 mm
Tip Diameter	d_a	21.75 h7 mm
Root Diameter	d_f	16.125 mm
Root Fillet Radius	r_fP	0.38-0.47 mm
Tip Fillet Radius	r_tip	0.08±0.03 mm
Active Face Width	b	10.5 mm
Tooth Thickness Tolerance	DIN 3967	h24 (0/-0.02 mm)
Bore Size	d_bore	9.98 H6 mm Ra<0.8 um
Maximum Teeth Runout	bore to teeth (F_r)	0.008 mm
Maximum Axial Runout	bore to faces	0.006 mm
Normal Tooth Thickness	s_n	2.145 (0/-0.02 mm)
Chordal Tooth Thickness	s_ny	2.141 (0/-0.02 mm)
Measuring Pin Diameter	d_D	2.250 mm
Measurement Over 2 Pins	M_d	22.269 (0/-0.039 mm)

MATERIAL, PROCESSES, TREATMENTS
4140 steel quenched & tempered to 28-32 HRC
CNC hob then profile grind teeth flanks to grade 6 (Ra<0.4 um)
Plasma / Ion Nitride treatment to 58-62 HRC 0.25-0.35mm case depth
White compound layer maximum 8 um

ORDER OF OPERATIONS
1. Quench and temper stock
2.a Hob and grind
2.b Break sharp edges 0.1mm
3. Nitride treatment

Dept.	Technical reference	Created by	Approved by
R&D	ISO 2768-mK	Bence Visztra 07/11/2026	Bence Visztra 08/11/2026
Scale 4:1 Units: mm		Document type: Part Manufacturing Drawing	Document status: Released for Quote
		Title: PINION STAGE 1 40K gear drive (15T M1.25)	DWG No.: DRV-40K-101
			Rev.: A / Date of issue: 08/11/2026 / Sheet: 1/1

3 gear idler drive for downhill skateboard wheels

Designed for small scale production, using additive manufacturing (SLM process). 3D printed prototype done, gears will be manufactured early next year, housing will be CNC milled by myself.

Supports ~14k RPM ~7 Nm input.
Designed to minimize tracking width.

